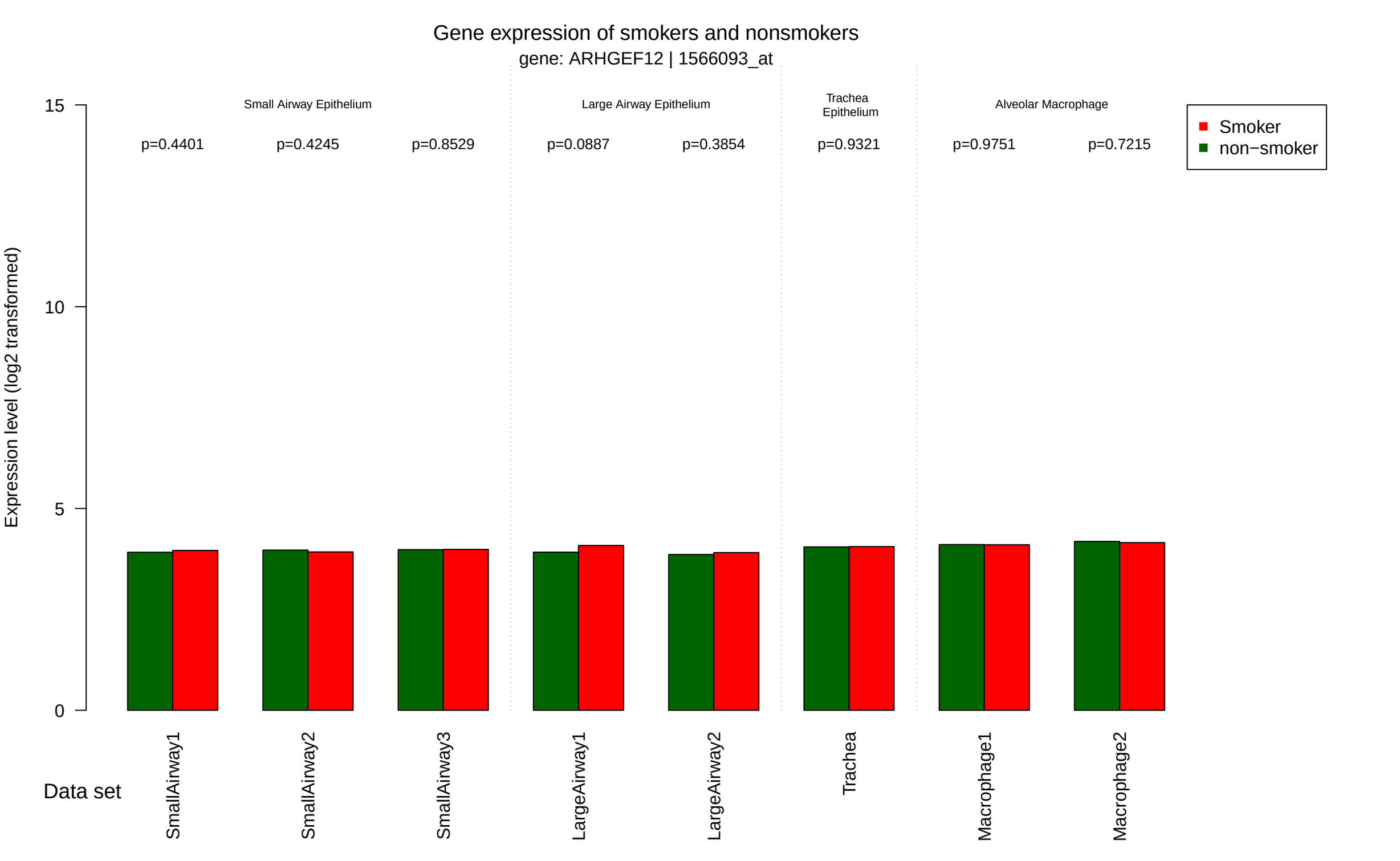
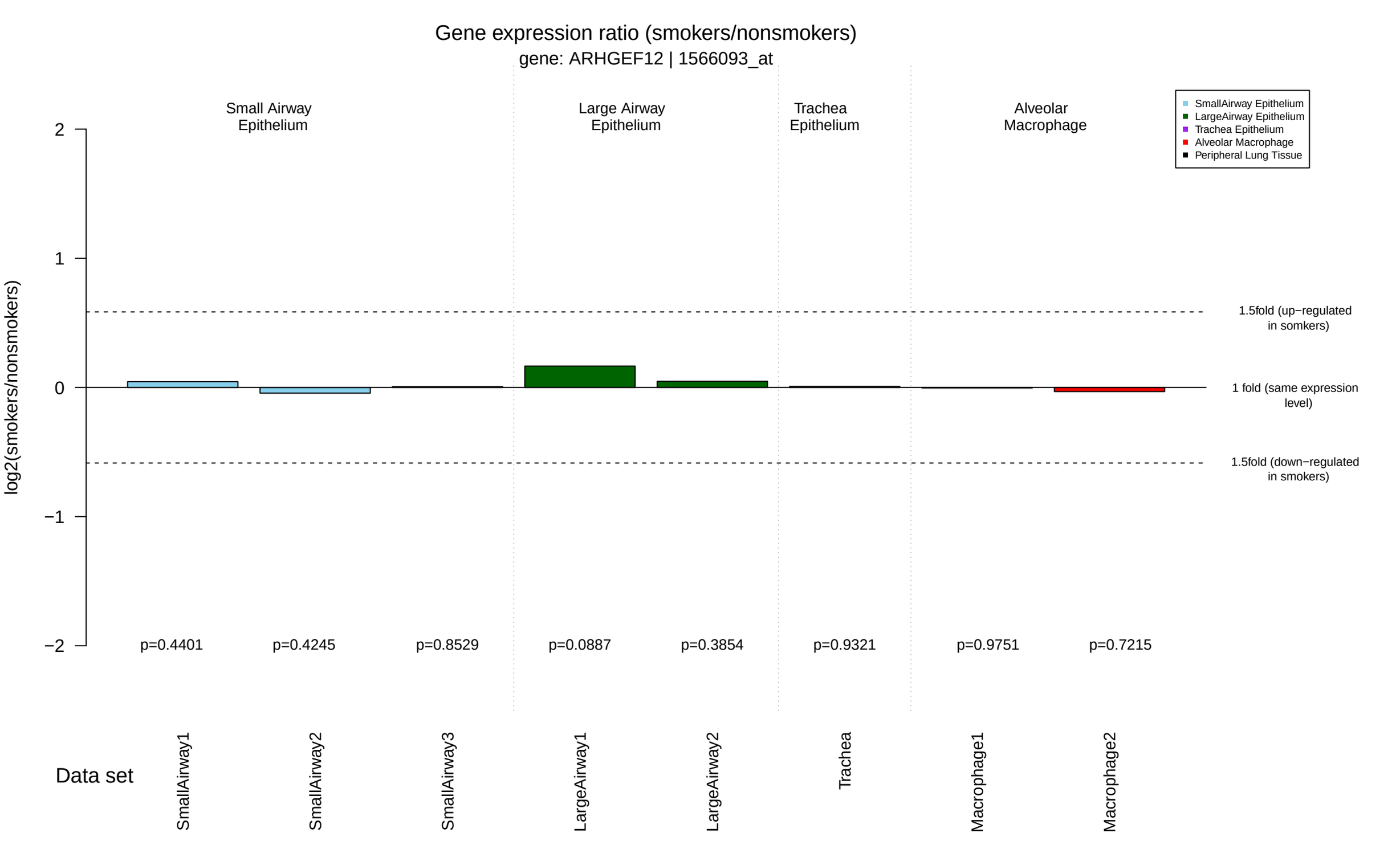
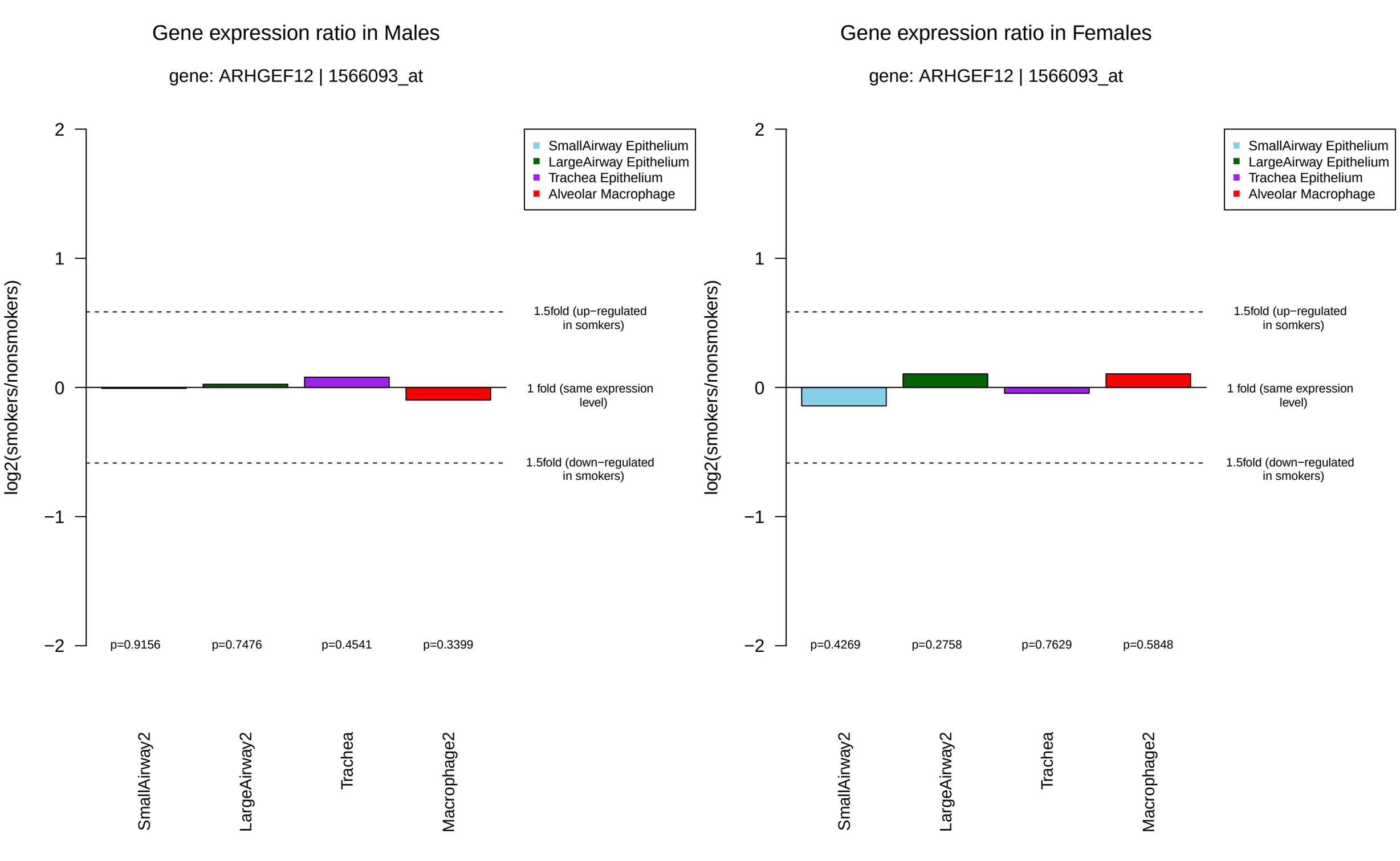
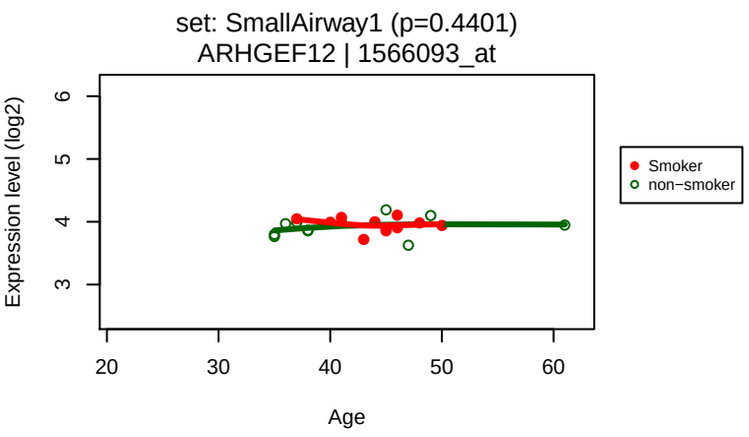
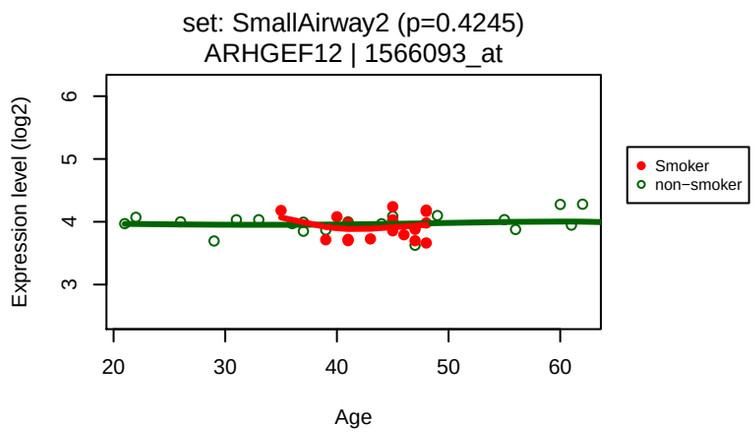
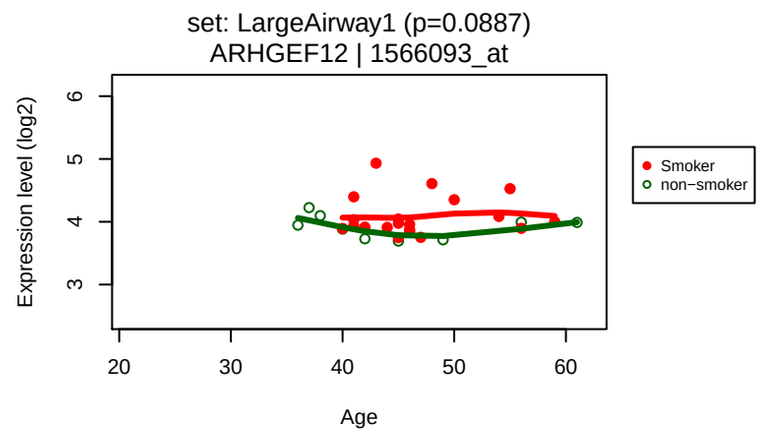

Gene expression of smokers and nonsmokers
gene: ARHGEF12 | 1566093_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4401
p=0.4245
p=0.8529
p=0.0887
p=0.3854
p=0.9321
p=0.9751
p=0.7215
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

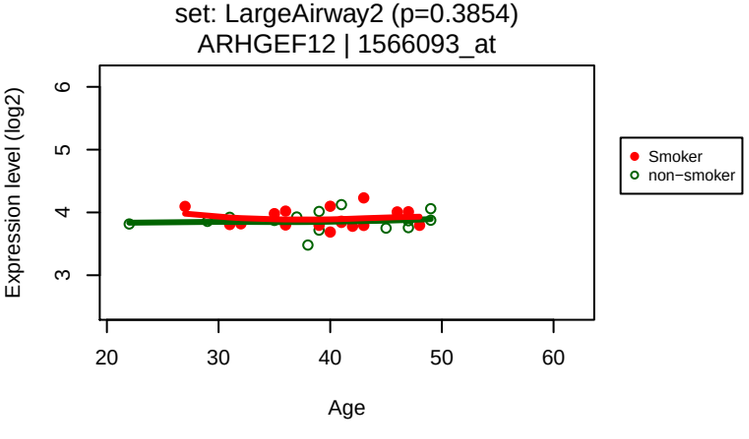

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF12 | 1566093_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4401
p=0.4245
p=0.8529
p=0.0887
p=0.3854
p=0.9321
p=0.9751
p=0.7215
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

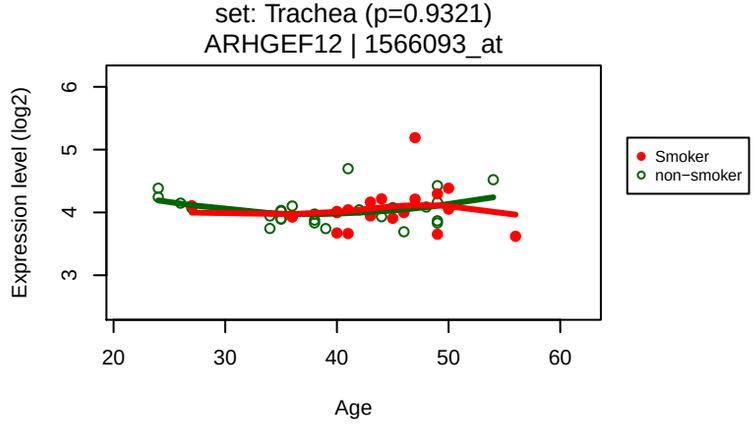

Gene expression ratio in Males
gene: ARHGEF12 | 1566093_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9156
p=0.7476
p=0.4541
p=0.3399
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF12 | 1566093_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4269
p=0.2758
p=0.7629
p=0.5848
SmallAirway2
LargeAirway2
Trachea
Macrophage2

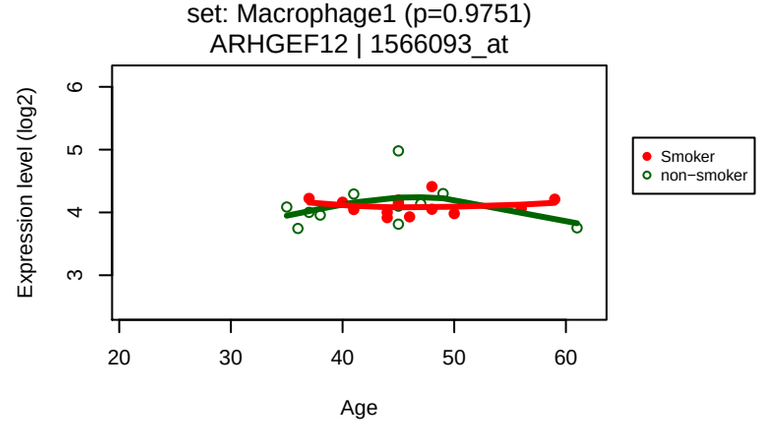

set: SmallAirway1 (p=0.4401)
ARHGEF12 | 1566093_at
Expression level (log2)
Age
Smoker
non−smoker

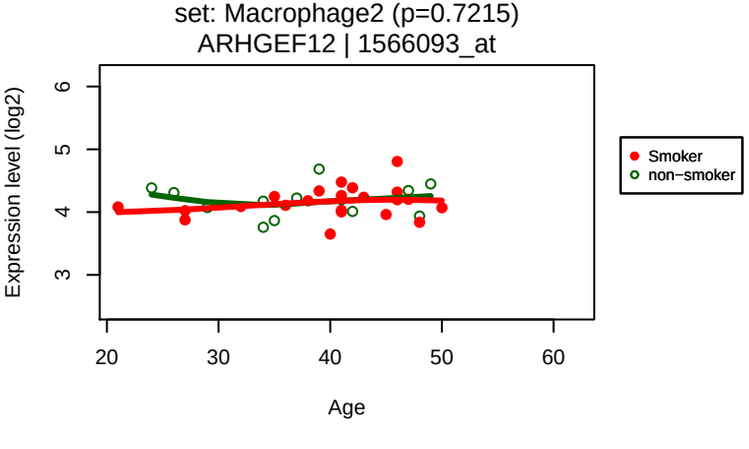

set: SmallAirway2 (p=0.4245)
ARHGEF12 | 1566093_at
Expression level (log2)
Age
Smoker
non−smoker

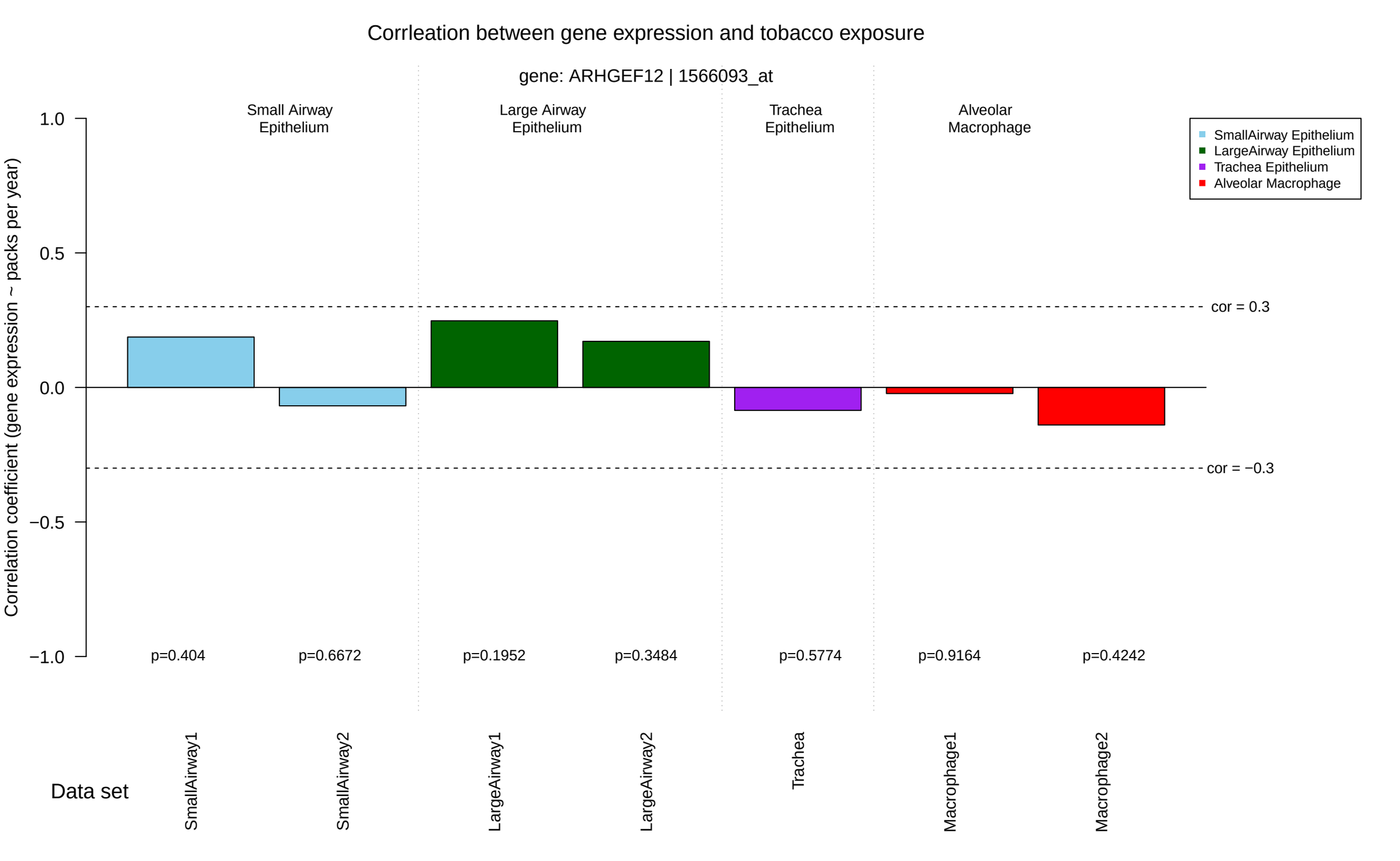

set: LargeAirway1 (p=0.0887)
ARHGEF12 | 1566093_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3854)
ARHGEF12 | 1566093_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9321)
ARHGEF12 | 1566093_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9751)
ARHGEF12 | 1566093_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7215)
ARHGEF12 | 1566093_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF12 | 1566093_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.404
p=0.6672
p=0.1952
p=0.3484
p=0.5774
p=0.9164
p=0.4242
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2